

⋮
⋮

Financial Econometrics I

Modeling Volatility II (Lecture 5)

Summer Semester 2025/2026 by Lukas Vacha

[join online lecture in MS Teams](#)

* If viewed in .pdf format - for full functionality use *Mathematica* 7 notebook (.nb) version of this .pdf

Conditional Heteroscedastic Models - II

Conditional heteroscedastic models are used for modeling σ_t^2 as an volatility estimate by "allowing" heteroscedasticity (time-variation) and capturing that dependence.

Discussion of the reading

IGARCH

Forecasting with GARCH

EWMA

GARCH-t

TAR-GARCH

EGARCH

Introduction to Long memory

Long memory in volatility

IGARCH - Integrated GARCH

Motivating example: two samples with different unconditional variance (SPX index):

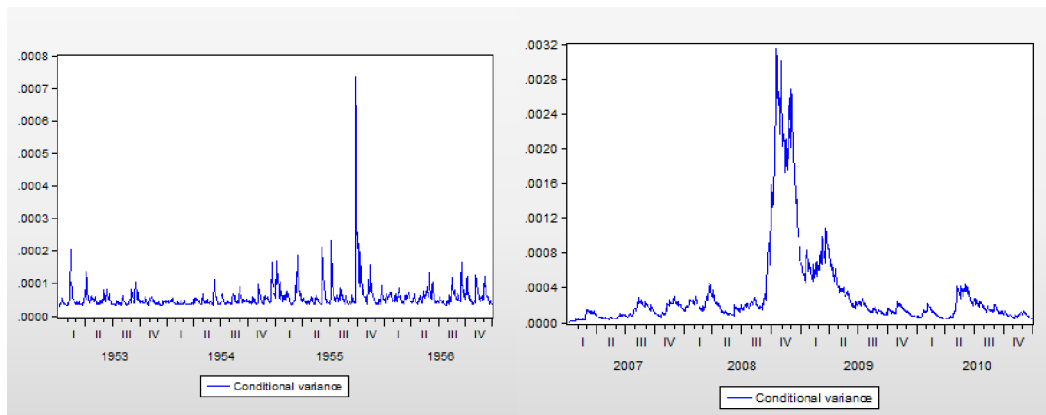
We can clearly observe different levels of GARCH model persistence (sum of α and β)

Sample 1: 1/02/1950 12/31/1953:

$$\sigma_t^2 = 0.000011 + 0.64 \sigma_{t-1}^2 + 0.14 a_{t-1}^2, \quad \text{t-dof} = 5.079.$$

Sample 2: 1/02/2007 12/31/2010:

$$\sigma_t^2 = 0.0000017 + 0.9007 \sigma_{t-1}^2 + 0.09989 a_{t-1}^2, \quad \text{t-dof} = 5.651.$$



Where do we need IGARCH?

IGARCH models are unit-root (integrated) GARCH models where $\alpha_1 + \beta_1 = 1$, their key feature is, that past squared shocks are persistent. The unconditional variance of a_t is not defined under the IGARCH(1,1) model specification.

The IGARCH effect might be caused by level shifts in volatility, i.e., can be spurious, (for example long samples. Similar to simple AR processes.) see Mikosch and Stărică (2004).

IGARCH(1,1) is defined as:

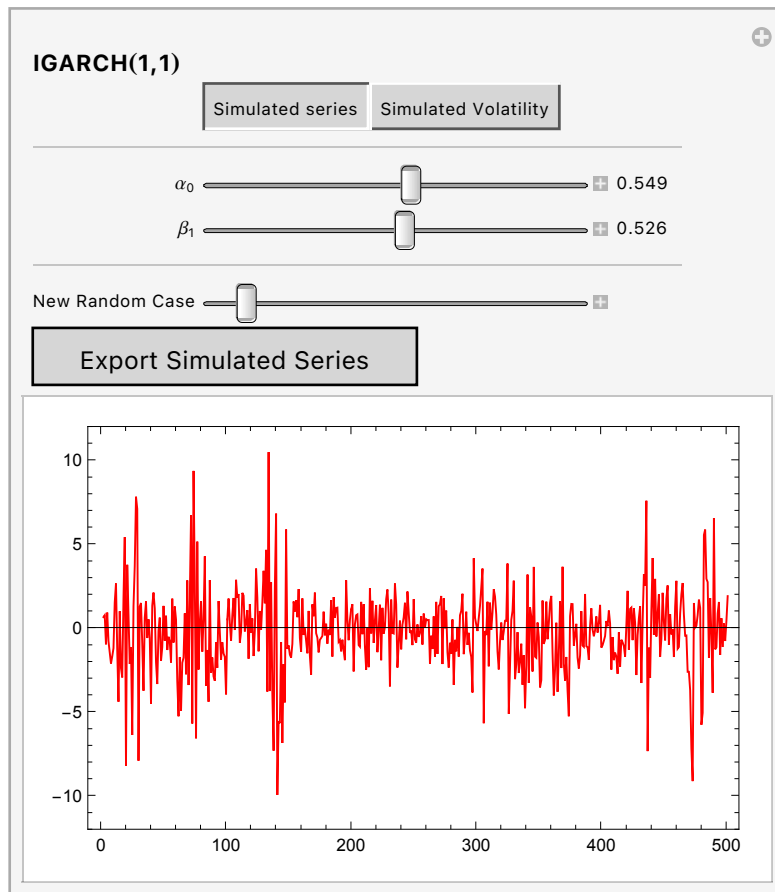
$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) a_{t-1}^2,$$

where $0 < \beta_1 < 1$

For $\alpha_0 = 0$, the IGARCH(1,1) $\rightarrow$ infinite EWMA.

Example IGARCH(1,1) artificial processes



Forecasting with GARCH

GARCH(1,1) is most common for financial applications. The volatility equation combines connection with past returns as well as autoregressive dependence.

$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

Forecasts of the GARCH can be obtained recursively at origin h , the 1-step ahead forecast of σ_{h+1}^2 is:

$$\sigma_{h+1}^2 = \alpha_0 + \alpha_1 a_h^2 + \beta_1 \sigma_h^2$$

For multi-forecast, we use $a_t^2 = \sigma_t^2 \epsilon_t^2$ and since $E(\epsilon_{h+1}^2 | F_h) = 1$, we have,

$$\begin{aligned} \sigma_{h+1}^2 &= \alpha_0 + (\alpha_1 + \beta_1) \sigma_h^2 \\ \sigma_{h+2}^2 &= \alpha_0 + (\alpha_1 + \beta_1) \sigma_{h+1}^2 \\ \sigma_{h+l+1}^2 &= \alpha_0 + (\alpha_1 + \beta_1) \sigma_{h+l}^2, \quad l > 1 \end{aligned}$$

$$\sigma_{h+l}^2 \rightarrow \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \text{as } l \rightarrow \infty.$$

Hence, the forecast converges to the **unconditional variance**.

Forecasting with the Exponential Moving Average (EWMA)

Alternatively, exponential moving average (EWMA) estimator of volatility can be used. It uses all data points, and recent observations carry larger weights. (weights are exponentially decreasing). It also has the effect of diminishing the problematic "Ghost Features".

An **m-period EWMA** with smooth constant λ is defined as:

$$\begin{aligned}\sigma_t^2 &= \frac{\sum_{\tau=1}^m \lambda^{\tau-1} r_{t-1-\tau}^2}{1+\lambda^1+\dots+\lambda^{m-1}} \quad \text{where } 0 < \lambda < 1 \\ \sigma_{t+1|t}^2 &= (1-\lambda) \sum_{\tau=0}^{\infty} \lambda^{\tau} r_{t-\tau}^2 \quad \text{as } m \rightarrow \infty; \\ \sigma_{t+1|t}^2 &= (1-\lambda) r_t^2 + \lambda(1-\lambda) \left(\sum_{\tau=1}^{\infty} \lambda^{\tau-1} r_{t-\tau}^2 \right) \\ \text{using the approximation } \sum_{\tau=1}^{\infty} \lambda^{\tau-1} &\simeq \frac{1}{1-\lambda} \quad \text{we get:}\end{aligned}$$

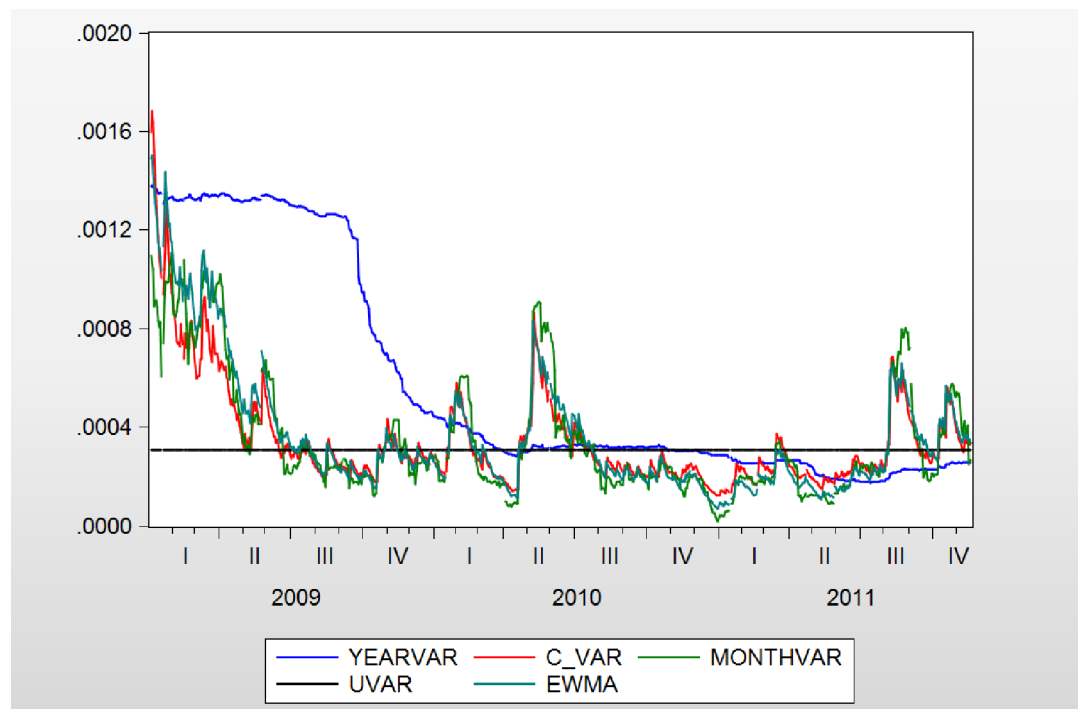
$$\sigma_{t+1|t}^2 = (1-\lambda) r_t^2 + \lambda \sigma_{t|t-1}^2,$$

where the initial value could be unconditional variance in a historical sample.
In most financial applications, $\lambda=0.94$ is used.

The EWMA model is equivalent to the IGARCH model, that is why the next day forecasts look very similar to forecasts based on GARCH type models.

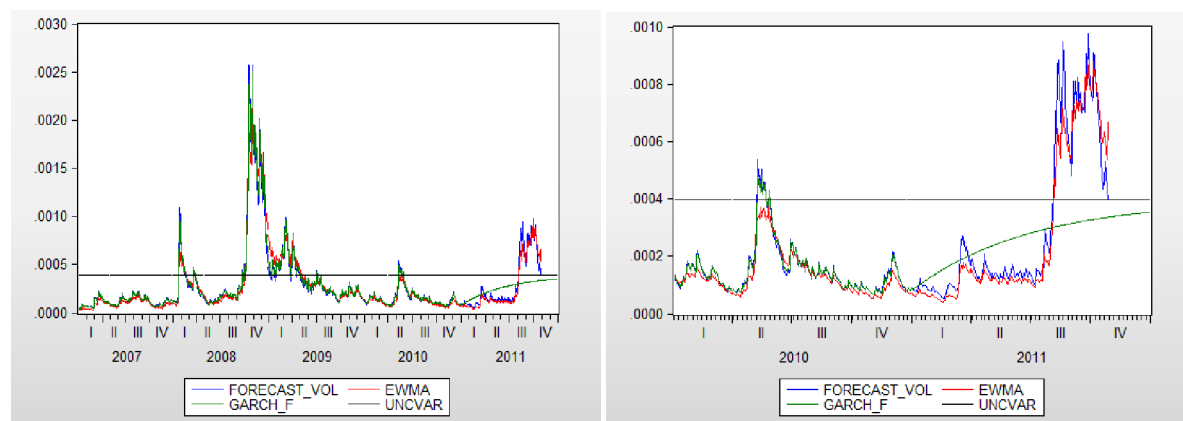
The EWMA ignores the long-run variance (unconditional), while the forecasts based on GARCH family models converge in the long-run to the unconditional variance of the sample.

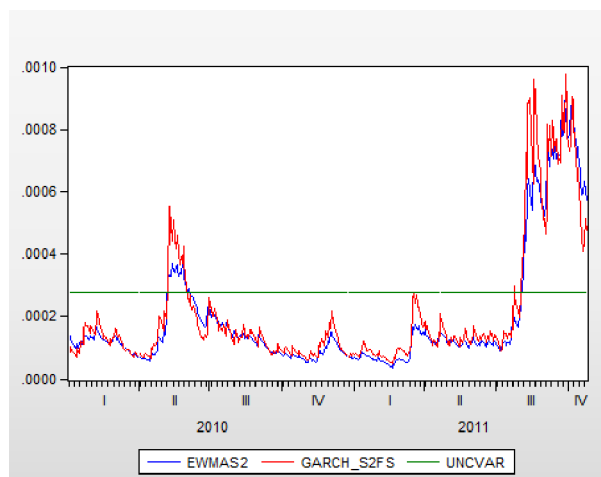
Example 1 month, 1 year volatility of Apple.



Example: Real financial market data -DAX index

Forecasts of variance for the DAX index. Two samples with different unconditional variance.





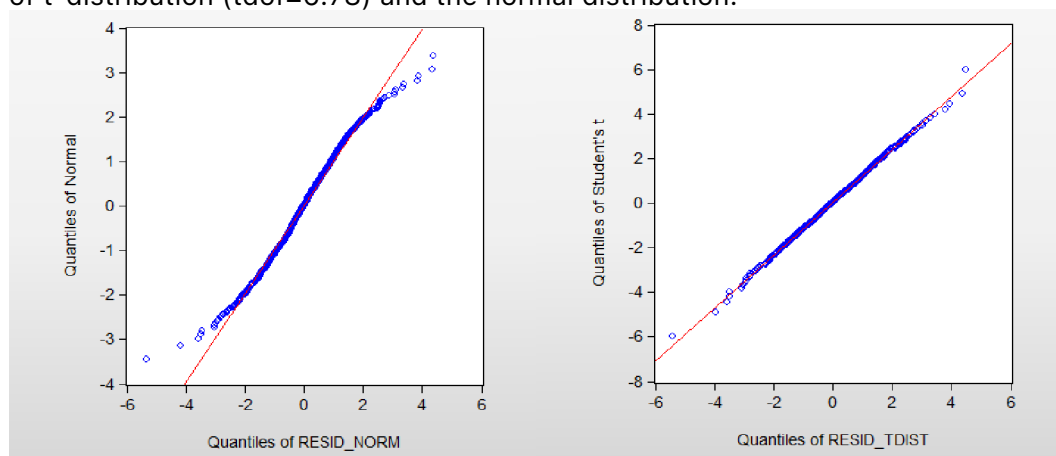
GARCH with t innovations

Are the GARCH residuals normally distributed? Usually the financial assets prices are fat-tailed, i.e. the distribution has more weight in the tails than a normal distribution. In general, there is a higher probability of large gain (or loss) than indicated by the normal distribution.

Instead of the Gaussian innovations we can use t-distributed innovations.

Example: with Q-Q plot

Apple (AAPL) daily closing prices GARCH(1,1), Q-Q plots of residuals against the quantiles of t-distribution (tdof=6.78) and the normal distribution:



Threshold Autoregressive (TAR)-GARCH Model

Threshold Autoregressive (TAR) model can be used to refine the model by allowing for asymmetric response in the (volatility) equation to the sign of shock. We can observe asymmetry in declining and rising patterns of examined time series. Model uses simple threshold to improve linear approximation. We demonstrate the idea on a simple two regime AR(1) model and then we proceed to threshold model in the variance equation.

Examples of Two Regime AR (1) model

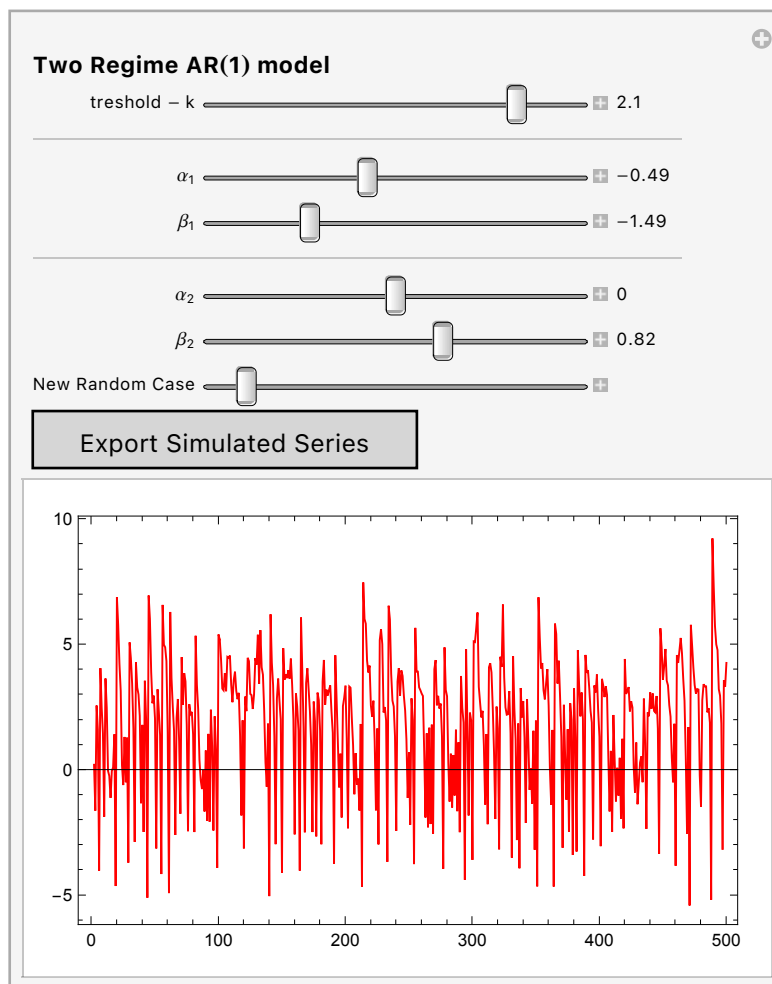
Two-Regime AR(1) model is represented by:

$$x_t = \begin{cases} \alpha_1 + \beta_1 x_{t-1} + \epsilon_t & x_{t-1} < k \\ \alpha_2 + \beta_2 x_{t-1} + \epsilon_t & x_{t-1} \geq k \end{cases}$$

Parameters are initially set to: $\alpha_1 = \alpha_2 = 0$, $\beta_1 = -1.5$ and $\beta_2 = 0.5$ to obtain following Two Regime AR(1) process:

$$x_t = \begin{cases} -1.5 x_{t-1} + \epsilon_t & x_{t-1} < 0 \\ 0.5 x_{t-1} + \epsilon_t & x_{t-1} \geq 0 \end{cases}$$

Note, that the process is stationary despite the coefficient -1.5 in the first regime. Series contains large upward jumps when it becomes negative (due to -1.5 coefficient), and there are more positive than negative ones. Model also contains no constant term, but $E(x_t)$ is not zero.



Example

Set parameters of previous example to: $\alpha_1 = \alpha_2 = 0$, $\beta_1 = 0.2$ and $\beta_2 = 0.9$ to obtain following Two Regime AR(1) process

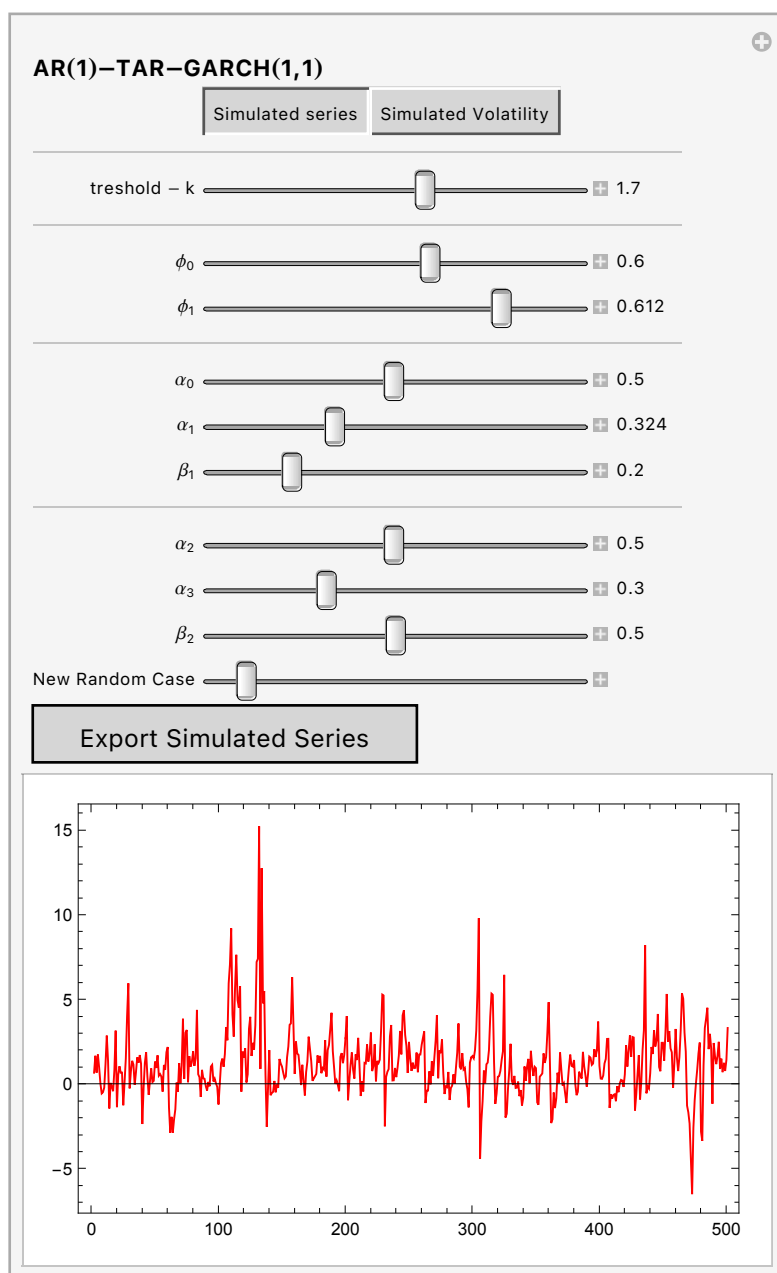
$$x_t = \begin{cases} 0.2 x_{t-1} + \epsilon_t & x_{t-1} < k \\ 0.9 x_{t-1} + \epsilon_t & x_{t-1} \geq k \end{cases}$$

and see the mixture of short-memory and long-memory processes

Example of application: AR(1)-TAR-GARCH(1,1)

TAR models can be used for capturing asymmetric response in volatility. If we find the series to follow AR(1)-GARCH(1,1) process, we can introduce TAR to model asymmetric response in volatility to shocks. Model is formalized:

$$\begin{aligned} r_t &= \phi_0 + \phi_1 r_{t-1} + a_t, \\ a_t &= \sigma_t \epsilon_t \\ \sigma_t^2 &= \begin{cases} \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 & a_{t-1} \leq 0 \\ \alpha_2 + \alpha_3 a_{t-1}^2 + \beta_2 \sigma_{t-1}^2 & a_{t-1} > 0 \end{cases} \end{aligned}$$



GJR-GARCH

GJR-GARCH is a simple version of threshold GARCH model. The volatility equation has a form:

$$\sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + \alpha_1 a_{t-1}^2 + \gamma I(a_{t-1} \leq 0) a_{t-1}^2$$

where I is the indicator function.

Empirical example

Apple (AAPL) daily closing prices sample: 9/1/2009 11/18/2011 (561 obs.)

Estimated volatility eq. (normal innovations):

$$\sigma_t^2 = 0.00005 + 0.665 \sigma_{t-1}^2 + 0.244 a_{t-1}^2 I(a_{t-1} \leq 0)$$

Estimated volatility eq. (t innovations): $\sigma_t^2 = 0.00004 + 0.71 \sigma_{t-1}^2 + 0.240 a_{t-1}^2 I(a_{t-1} \leq 0)$

where $I(a_{t-1} < 0) = 1$ if $a_{t-1} < 0$.

EGARCH Model

The EGARCH (Nelson, 1991) model be also used for capturing asymmetric response in volatility - the leverage effect. The EGARCH(m,s) can be defined as:

$$a_t = \sigma_t \epsilon_t$$

$$\ln(\sigma_t^2) = \alpha_0 + \frac{1 + \alpha_1 L + \dots + \alpha_m L^m}{1 - \beta_1 L - \dots - \beta_s L^s} g(\epsilon_{t-1})$$

$$g(\epsilon_t) = \theta \epsilon_t + \gamma [|\epsilon_t| - E|\epsilon_t|]$$

where $\alpha_0 > 0$, $\theta, \gamma \in \mathbb{R}$ are real constants. For the standard Gaussian random variable ϵ_t , $E|\epsilon_t| = \sqrt{2/\pi}$.

Function $g(\cdot)$ captures the asymmetric relations between stock returns and volatility. It is both function of magnitude and sign of ϵ_t . Each component of $g(\epsilon_t)$ has mean zero. Thus, $g(\epsilon_t)$ is a zero mean i.i.d. random sequence.

When ϵ_t is positive (negative) $g(\epsilon_t)$ is linear in ϵ_t with slope $\theta + \gamma$ ($\theta - \gamma$). Thus, $g(\epsilon_t)$ allows for asymmetrical response of conditional variance (σ_t^2) to positive or negative returns.

Example EGARCH(0,1)

$$(1 - \beta_1 L) \ln(\sigma_t^2) = \alpha_0 + g(\epsilon_{t-1})$$

Long memory - motivation

Several disciplines came to realizations that naturally occurring phenomena (river flow, atmospheric patterns, telecommunications, economics, etc) exhibit correlations that do not decay at a sufficiently fast rate.

It is difficult to distinguish between $I(1)$ and $I(0)$. Thus, knife-edge distinction between $I(0)$ and $I(1)$ processes can be far too restrictive.

Volatility of asset prices can be successfully approximated by long memory models.

Significant correlations in observations are separated by great periods of time.

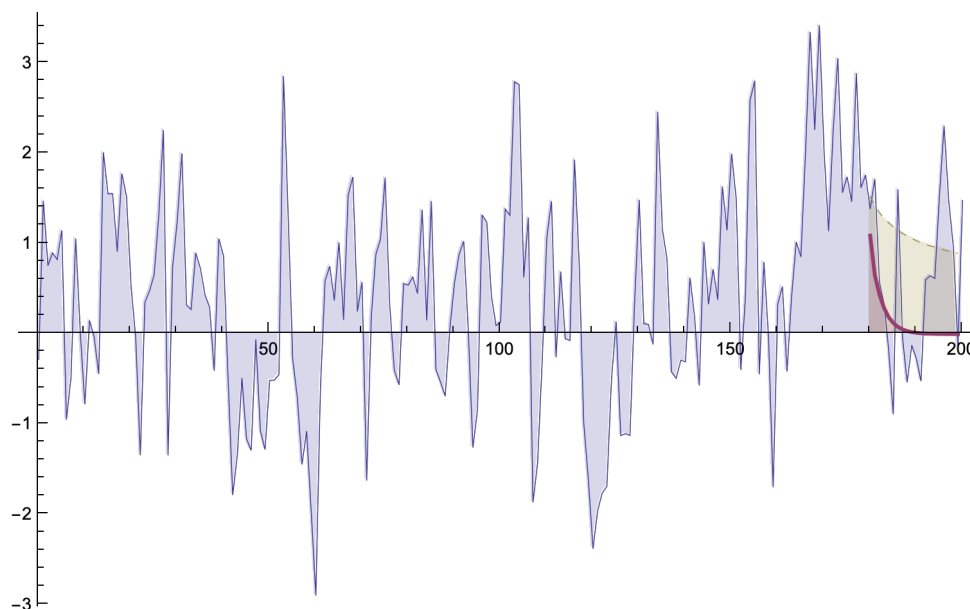
Occurrence of jumps

Long memory time series require different approaches to modeling than the short memory time series.

Application to fin. markets: Ding, Zhuanxin, Clive WJ Granger, and Robert F. Engle. "A long memory property of stock market returns and a new model." *Journal of empirical finance* 1.1 (1993): 83-106.

Example:

prediction of long memory process: Red line is prediction based on AR(3) and the yellow one is appropriate long memory process.



Concept of Long memory

There are various definitions of long memory! We focus on a covariance structure of the process:

Definition: Let r_t be a stationary process for which the following holds. There exists a real number $\alpha \in (0, 1)$ and a constant $c_\rho > 0$ such that

$$\lim_{k \rightarrow \infty} \frac{\rho(k)}{[c_\rho k^{-\alpha}]} = 1.$$

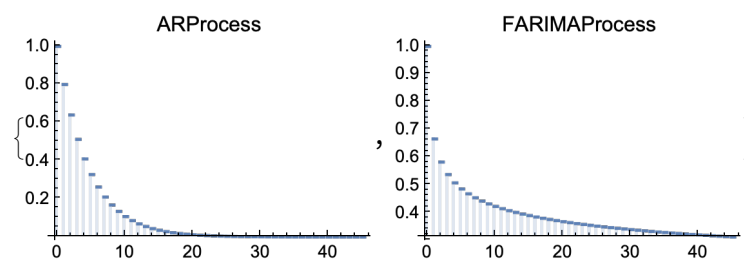
Then r_t is called a stationary process with long memory or long range dependence or strong dependence, or a stationary process with slowly decaying or long range correlations (Beran 1994).

Further we use memory parameter $d=(1-\alpha)/2$.

In case $d=0$, ($\alpha=1$), then r_t is a random process with exponentially decaying ACF.

Example ACF of short and long memory process:

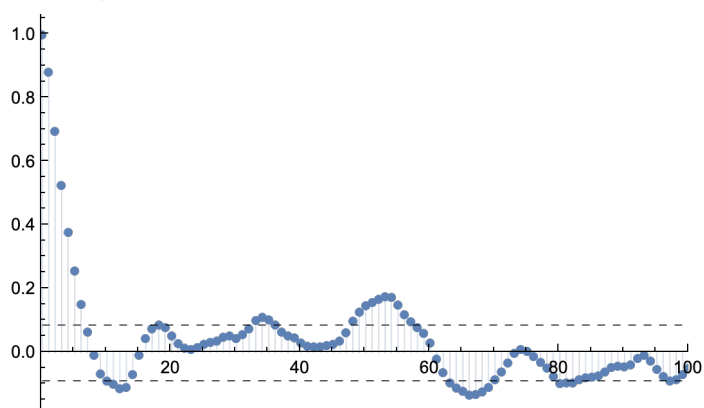
```
DiscretePlot[CorrelationFunction[FARIMAProcess[ {.1}, .49, {.3}, 1], h],
 {h, 0, 100}, ExtentSize -> 1 / 1.5, PlotRange -> All]
```



Long memory: $0 < d < \frac{1}{2}$

Problems with long-memory detection: only the rate of convergence is defined, not the absolute size.

Example: ACF plot with white-noise confidence bands



Fractional ARIMA

ARIMA model were introduced by Box and Jenkins in 1970.

Fractional ARIMA (ARFIMA, FARIMA) models are natural extension of the classic ARIMA.

ARMA (p, q):

$$r_t = \phi_0 + \sum_{i=1}^p \phi_i r_{t-i} + \epsilon_t - \sum_{i=1}^q \theta_i \epsilon_{t-i}$$

or

$$(1 - \phi_1 L - \dots - \phi_p L^p) r_t = \phi_0 + (1 + \theta_1 L + \dots + \theta_q L^q) \epsilon_t$$

ARIMA (p, d, q):

Series are said to be ARIMA(p,1,q), when the process $r_t - r_{t-1} = (1 - L) r_t$ follows a stationary and invertible ARMA(p,q). Parameter d is equal to 1, and means, that ARMA(p,q) series are differenced once (parameter d is an integer, $d \geq 0$).

$$\phi(L) (1 - L)^d r_t = \theta(L) \epsilon_t$$

ARFIMA generalize ARIMA by allowing d to assume any real value.

Definition ARFIMA: Let r_t be a stationary process such that

$$(1 - L)^d \phi(L) r_t = \theta(L) \epsilon_t$$

for some $-\frac{1}{2} < d < \frac{1}{2}$. Then r_t is called a fractional ARIMA (p,d,q) process.

The parameter d determines the long-term behavior, parameters p and q allow for more flexible modelling of the short-range properties.

For $(1 - L)^d$, where d is any real number we can write:

$$(1 - L)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k L^k \quad \left(d = H - \frac{1}{2}\right),$$

$$\binom{d}{k} = \frac{d!}{k! (d-k)!} = \frac{\Gamma(d+1)}{\Gamma(k+1) \Gamma(d-k+1)},$$

where $\Gamma(\cdot)$ is Euler's gamma function.

Thus, when we apply the fractional differencing operator (which is in fact a filter) we obtain:

$$Z_t = (1 - L)^d r_t = \phi(L)^{-1} \theta(L) \epsilon_t$$

where Z_t is the ordinary ARMA(p,q) process.

Example: Approximation of the ARFIMA process with the MA and AR process

```
MAProcess[FARIMAProcess[{0}, .4, {.0}, 1], 20]
```

```

MAProcess[{0.4, 0.28, 0.224, 0.1904, 0.167552, 0.150797, 0.137871,
  0.127531, 0.119029, 0.111887, 0.105784, 0.100495, 0.0958568, 0.0917487,
  0.0880787, 0.0847758, 0.0817837, 0.0790576, 0.076561, 0.0742642}, 1]

ARProcess[FARIMAProcess[{0}, .4, {.0}, 1], 20]

ARProcess[{0.4, 0.12, 0.064, 0.0416, 0.029952, 0.0229632, 0.0183706, 0.0151557,
  0.0127982, 0.0110064, 0.0096056, 0.00848495, 0.00757118, 0.00681406,
  0.00617808, 0.0056375, 0.00517324, 0.00477087, 0.00441934, 0.00410998}, 1]

```

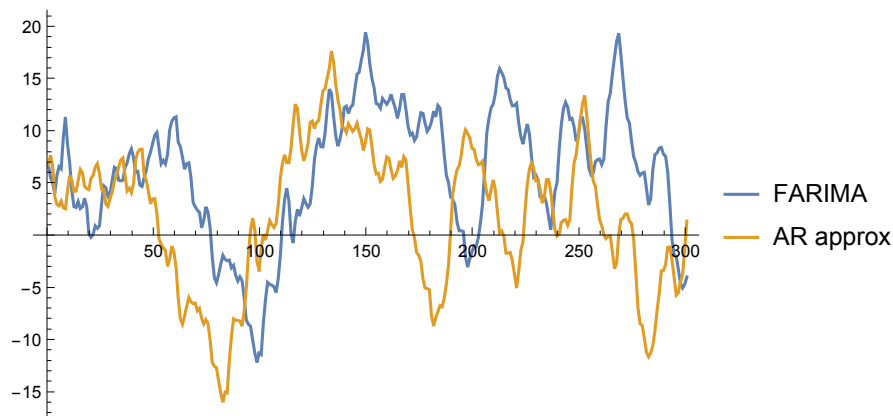
Example: Approximation of ARFIMA by an AR process

```

proc = FARIMAProcess[{.8}, .4, {.5}, 1];
aproc = ARProcess[proc, 15]
SeedRandom[13]; sample = RandomFunction[proc, {300}];
SeedRandom[13]; asample = RandomFunction[aproc, {300}];
ListLinePlot[{sample, asample}, PlotLegends -> {"FARIMA", "AR approx"}]

ARProcess[{1.7, -1.05, 0.493, -0.2561, 0.124722,
  -0.0633594, 0.0316797, -0.0153806, 0.00836388, -0.00341405,
  0.00250749, -0.000453279, 0.00100987, 0.000252185, 0.000600741}, 1]

```



ACF for ARFIMA (0, d, 0)

ARFIMA (0, d, 0): $(1 - L)^d r_t = \varepsilon_t$

The autocorrelations of such a process decline at a hyperbolic rate to zero ($0 < d < \frac{1}{2}$), a much slower rate of decay than the exponential decay of the ARMA process. The sum of the absolute values of the autocovariance function is infinite; this is the case of long memory or long range dependence (Hosking 1981).

Let us assume $\mu = E(r_t) = 0$, ε_t is a white noise and the process is stationary and invertible, then the autocovariance is defined as

$$\gamma(k) = \sigma^2 \frac{(-1)^k \Gamma(1-2d)}{\Gamma(k-d+1) \Gamma(1-k-d)}$$

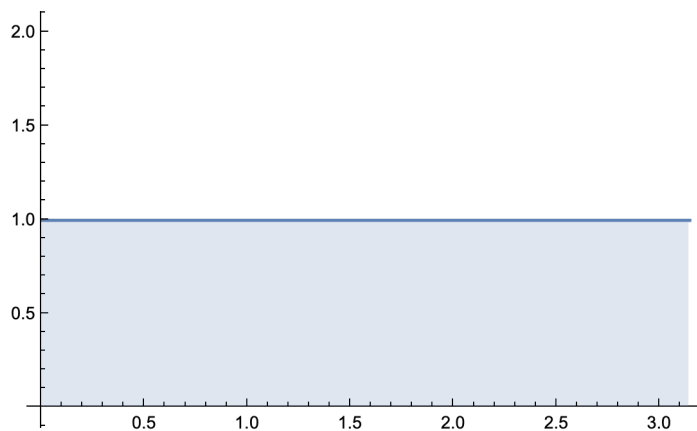
and the autocorrelation

$$\rho(k) = \frac{\Gamma(1-d) \Gamma(k+d)}{\Gamma(d) \Gamma(k+1-d)},$$

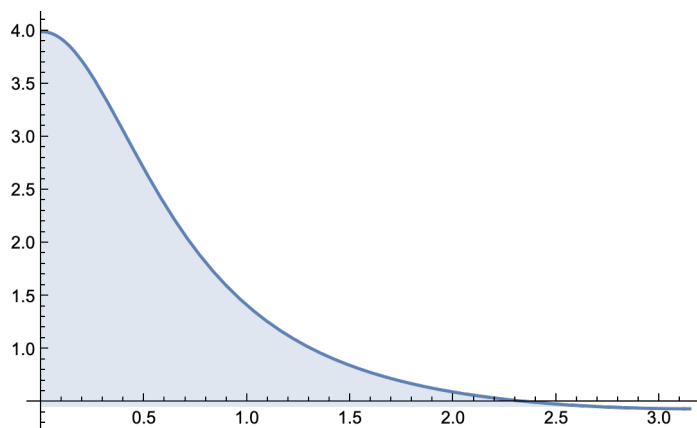
$$\rho(k) \sim \frac{\Gamma(1-d)}{\Gamma(d)} |k|^{2d-1} \quad |k| \rightarrow \infty.$$

Example: Spectral density of white noise, AR and ARFIMA (optional)

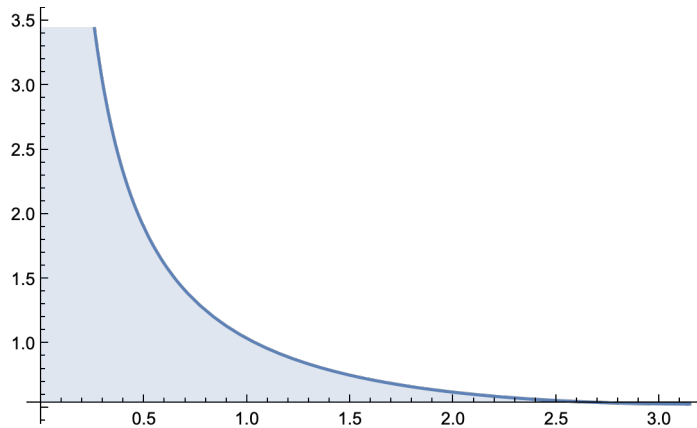
```
Plot[PowerSpectralDensity[FARIMAProcess[ {.0}, .0, {0}, 1], w],
{w, 0, π}, Filling → Axis]
```



```
Plot[PowerSpectralDensity[ARProcess[ {.5}, 1], w],
{w, 0, π}, Filling → Axis, PlotRange → All]
```



```
Plot[PowerSpectralDensity[FARIMAProcess[ {.0}, .45, {0}, 1], w],
{w, 0,  $\pi$ }, Filling -> Axis(*, PlotRange -> All*)]
```



GARCH as an ARMA process

Letting $a_t = r_t - \mu_t$ be the mean-corrected log return, the GARCH(m, s) model is defined as:

$$\begin{aligned} a_t &= \sigma_t \epsilon_t, \\ \sigma_t^2 &= \alpha_0 + \sum_{i=1}^m \alpha_i a_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2, \end{aligned}$$

where $\{\epsilon_t\}$ is i.i.d random variables with zero mean and variance 1, $\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$.

The GARCH(m, s) model can be also written with polynomials as:

$$\sigma_t^2 = \alpha_0 + \alpha(L) a_t^2 + \beta(L) \sigma_t^2 \quad (*)$$

where L is the lag operator, $\alpha(L) = \sum_{i=1}^m \alpha_i L^i$ and $\beta(L) = \sum_{j=1}^s \beta_j L^j$.

Defining $v_t = a_t^2 - \sigma_t^2$ we can rewrite (*) as an ARMA (p, m) process:

$$[1 - \alpha(L) - \beta(L)] a_t^2 = \alpha_0 + [1 - \beta(L)] v_t$$

where $p = \max\{m, s\}$.

If we allow for the presence of a unit root in $[1 - \alpha(L) - \beta(L)]$, we can define the IGARCH(m, s) model as:

$$(1 - L) \phi(L) a_t^2 = \alpha_0 + [1 - \beta(L)] v_t.$$

where $\phi(L) = \sum_{i=1}^{s-1} \phi_i L^i$ is of order $s-1$.

FIGARCH Model

Baillie (1996, JoE) and Bollerslev and Mikkelsen (1996) introduced the FIGARCH process, generalizing the GARCH to allow for persistence in the conditional variance.

If we compare the IGARCH variance equation with the ARFIMA model

$(1-L)^d \phi(L) X_t = \theta(L) \varepsilon_t$, we see that we can simply add the long memory term to the GARCH model and define the FIGARCH(m,d,s) model as:

$$\phi(L) (1-L)^d a_t^2 = \alpha_0 + [1 - \beta(L)] v_t.$$

where $\phi(L) = \sum_{i=1}^{s-1} \phi_i L^i$ is of order s-1.

FIEGARCH Model

Fractionally Integrated EGARCH (FIEGARCH) model was proposed by Bollerslev and Mikkelsen (1996). The model captures simultaneously with the leverage effect and long-memory.

Let us defines the FIEGARCH(m,d,s) model as

$$a_t = \sigma_t \epsilon_t$$

$$\ln(\sigma_t^2) = \alpha_0 + \frac{1+\alpha_1 L + \dots + \alpha_s L^s}{1-\beta_1 L - \dots - \beta_m L^m} (1-L)^{-d} g(\epsilon_{t-1})$$

where $g(\epsilon_t)$ is defined as:

$$g(\epsilon_t) = \theta \epsilon_t + \gamma[|\epsilon_t| - E|\epsilon_t|].$$

The term $(1-L)^{-d}$ is the long memory component with the long memory parameter $d \in (0, 1/2)$.

When $d = 0$, the short-memory EGARCH model of Nelson (1991) is obtained.

In its simplest form, the FIEGARCH(1,d,0) model is given by

$$(1 - \beta_1 L) (1 - L)^d \ln(\sigma_t^2) = \alpha_0 + g(\epsilon_{t-1})$$